



Fig. 2: Simulation on eSim (Image source: esim.fossee.in)

field-effect transistor (JFET), metal-oxide semiconductor field-effect transistor (MOSFET), magnetic core and IGBT.

Subcircuit Builder. Its main function is to create the sub-circuit for a component. The sub-circuit so created can be used in other circuits as well. Subcircuit Builder module has a feature to define new components like op-amps and IC 555. Users can also edit existing sub-circuits with this module.

Simulation

Built circuit layouts need to be verified for their validity, integrity and efficiency. Simulation software serves this purpose. eSim software includes a group of simulation tools to convert and visualise the circuit layout created using KiCad packages and Model Builder.

KiCad to Ngspice converter. The netlist generated by schematic editors needs to be converted into a Ngspice-compatible format. Developed in-house by the eSim team, the KiCad-to-Ngspice converter module helps to achieve that along with analysis parameters and source details. You can also add or edit device models and subcircuits present in the circuit schematic.

Simulation with Ngspice and NGHDL. Ngspice is a popular cir-

cuit simulator tool integrated into eSim platform. It can analyse non-linear DC, non-linear transient and linear AC lines. It supports almost all essential components including inductors, resistors, capacitors, individual current and voltage sources, transmission lines (both lossless as well as lossy), switches and distributed resistor-capacitor lines. It also supports all types of models provided by Model Builder. After the schematic and model are selected and converted, Ngspice displays the analysis in a graphical user interface.

NGHDL is an add-on simulation suite that eSim uses for mixed-mode circuit simulation using simulation tool GHDL.

OpenModelica. This is a modeling and simulation tool with two modules: OMEdit, an integrated development environment (IDE) for the modeling and simulation logic, and OMOptim, another IDE for optimising the simulation. It incorporates Modelica programming language.

Working with eSim

The eSim platform is quite user-friendly, and supports Windows (7, 8, 10) and Ubuntu (12.04) operating systems. The main user interface (UI) is quite familiar, with menu

bar, tool bar, dock area and console, project explorer and workspace. The toolbar allows users to access eSim modules as required.

Clicking 'Schematic Editor' button in the toolbar takes the platform to EESchema. EESchema has a series of toolbars that enlist the components and functionalities, which can be picked and placed on the workspace to create the circuit schematic. There are options of 'Footprint Editor,' which opens the CvPCB tool, and a layout editor, which opens the PCBnew tool. These steps create the netlist and the PCB layout.

The 'Convert KiCad to NgSpice' option enables format conversion. This section packs five tabs: Analysis, which sets the different parameters for analysing the circuit; Source Details, a dynamic tab scaled as per the number of power sources in the circuit; Device Modeling, which opens Model Builder; Subcircuits, which allows users to choose or create sub-circuits for components; and Ngspice Modeling, which opens the in-built Ngspice models. All this data needs to be entered. Once done, the conversion is processed. 'Simulation' button displays a Python plot of the circuit on a graphical user interface. The Ngspice-to-Modelica converter translates the simulated circuit in Modelica language, which is fed into the OMOptim for circuit optimisation.

The eSim library contains a comprehensive list of components and models. Users can also upload external library files to eSim.

Summing up

Developed under the Free and Open Source Software for Education (FOSS-SEE) initiative, eSim can be a decent alternative to commercial EDA solutions like OrCAD and HSPICE. The integration of PCB design and simulation in one package makes it a strong recommendation for basic- and medium-level users. **EFY**